University of St. La Salle
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HANDAVis
Stay Handa. See Risks Clearly.

HANDAVIS
PROJECT DOCUMENTATION

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BSIT 2B

Website Concept / Short Summary

HANDAVis is a web-based disaster monitoring and response system that includes users, barangay staff, responders, and admin in a single platform. It allows real time reporting, SOS alert, live map routing, emergency coordination and many more. It makes sure that data is verified, monitored, and updated across all roles. The entire concept aims to improve response time, disaster monitoring, and community safety.

Introduction

In times of disaster, accurate, fast, and reliable communication is important to ensure everyone's safety in the community. Many areas in Western Visayas experience delays in reporting incidents, some also lack real time updates, and a very poor coordination between the residents and the responders. These challenges can increase the risk of everyone during times of emergencies.

HANDAVis is a web-based system that improves and aims disaster monitoring and response. It provides a platform where users will be able to report hazards, receive important regional or barangay updates, and send SOS alerts. Simultaneously, the responders, barangay officials, and admin can manage, verify, and respond to incidents efficiently.

By integrating real-time data, notifications, mapping features, and communication tools. HANDAVis improves coordination and proper handling between all involved roles. Thus allowing faster decision making, better accurate information and flow, more efficient and effective emergency response time, and many more. The system aims to reduce the risk, improves the response time, and lastly to promote the safety of the community during disaster situations.

System Overview

User Roles

- **User** - Able to report hazards, receive alerts, send SOS, and route destinations.
- **Barangay** - Verifies reports, post alerts, and manage local incidents.
- **Responders** - Handles emergency cases and also updates the status.
- **Administrator** - Manages or monitors all systems, manages users, and oversees the entire operation.

Main Features

- Real time alert system
- Live mapping for disasters
- Hazard Reporting
- Evacuation Center Information and Routing
- In-app and SMS notifications
- Ai Assistant

The system contains four main users: User, Barangay, Responders, and Admin. Each of these roles has specific responsibilities and functionalities. These will allow them to interact with the system based on their access levels. Even though they have different functions and interfaces, all of these roles are connected into a single database where it ensures all data like reports and alerts are directly shared and also updated in real time.

HANDAVis also includes these main features such as Real time alert system, Live mapping for disasters, Hazard Reporting, Evacuation Center Information and Routing, In-app and SMS notifications, Ai Assistant. These features work together hand in hand to provide proper and efficient coordination, response, and communication in the times of disasters or emergencies.

System Architecture

This section outlines the core workflows of the system. This includes the report flow, SOS flow, and alert flow. These functions demonstrate how data is being processed, verified, escalated, and also displayed across the platform.

1. Report Flow

- Users submit reports.
- Barangay verifies the reports.
- If a report is urgent, it will be directed to Responders.
- Admin Monitors the reports.
- Status updates will go back to the User.

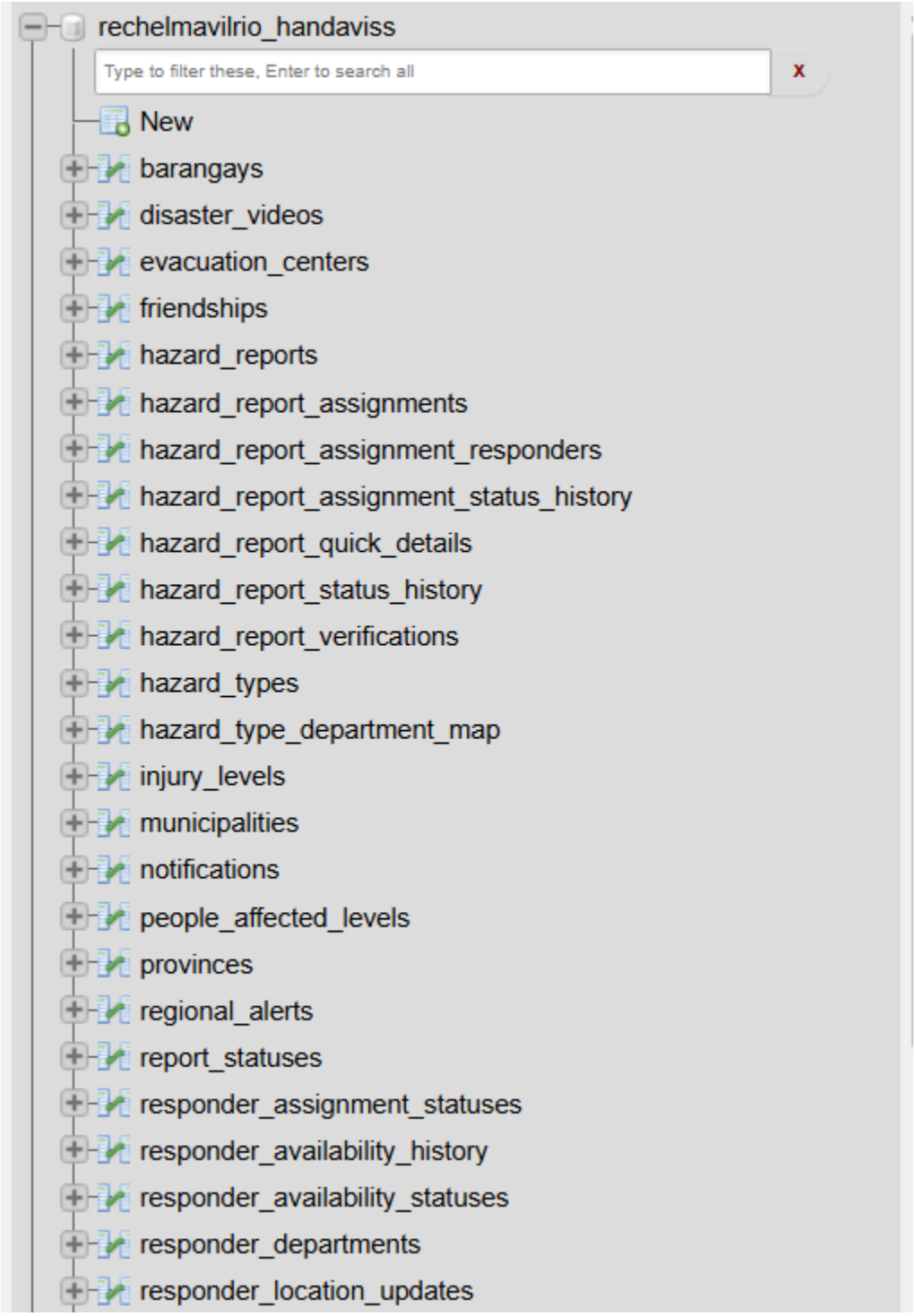
2. SOS Flow

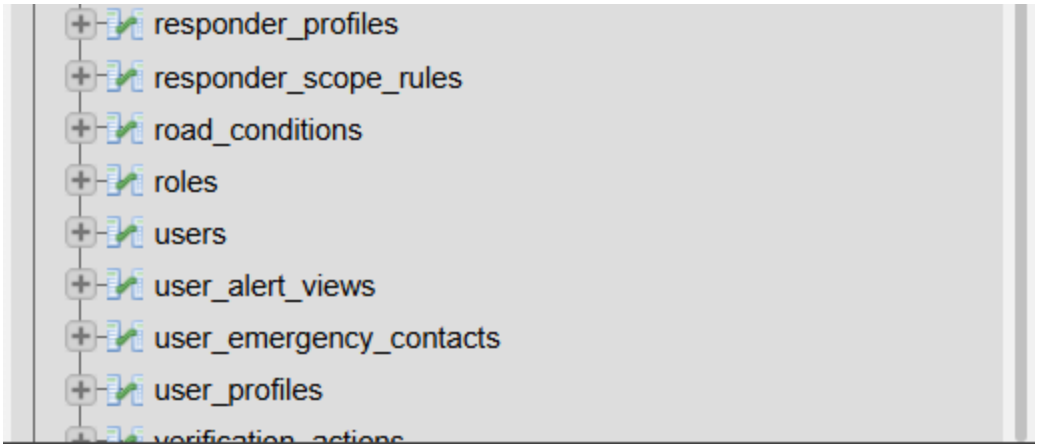
- Users press or send SOS.
- Barangay and/or Responders will be notified.
- Responders update progress.
- Admin logs everything.
- Users will see the updates once everything is processed.
- SMS will be sent to their respective mobile numbers.

3. Alert Flow

- Barangay and/or Admin creates an alert (Barangay/Regional)
- Alerts will be displayed in the User dashboard (User Alerts)

Database Design





HANDAVis database is designed to support a centralized monitoring system for users hazard reports, alerts, responders coordinations, and other location based data. The database is divided into several groups to ensure efficient data handling and real time updates.

User and Role Management

Handles accounts important personal data and information.

- users
- roles
- user_profile
- user_emergency_contacts
- friendships

Hazard Report Management

The core system handles the entire flow of reports.

- hazard_reports
- hazard_report_quick_details
- hazard_report_status_history
- hazard_report_verification
- hazard_report_assignments
- hazard_report_assignment_responders
- hazard_report_assignment_status_history

Alert System
Manages alerts in the website.
- regional_alerts
- user_alert_views

Responder Management
Handles emergency personnel.
- responder_profiles
- responder_departments
- responder_scope_rules
- responder_availability_statuses
- responder_availability_history
- responder_assignment_statuses
- responder_locaton_updates

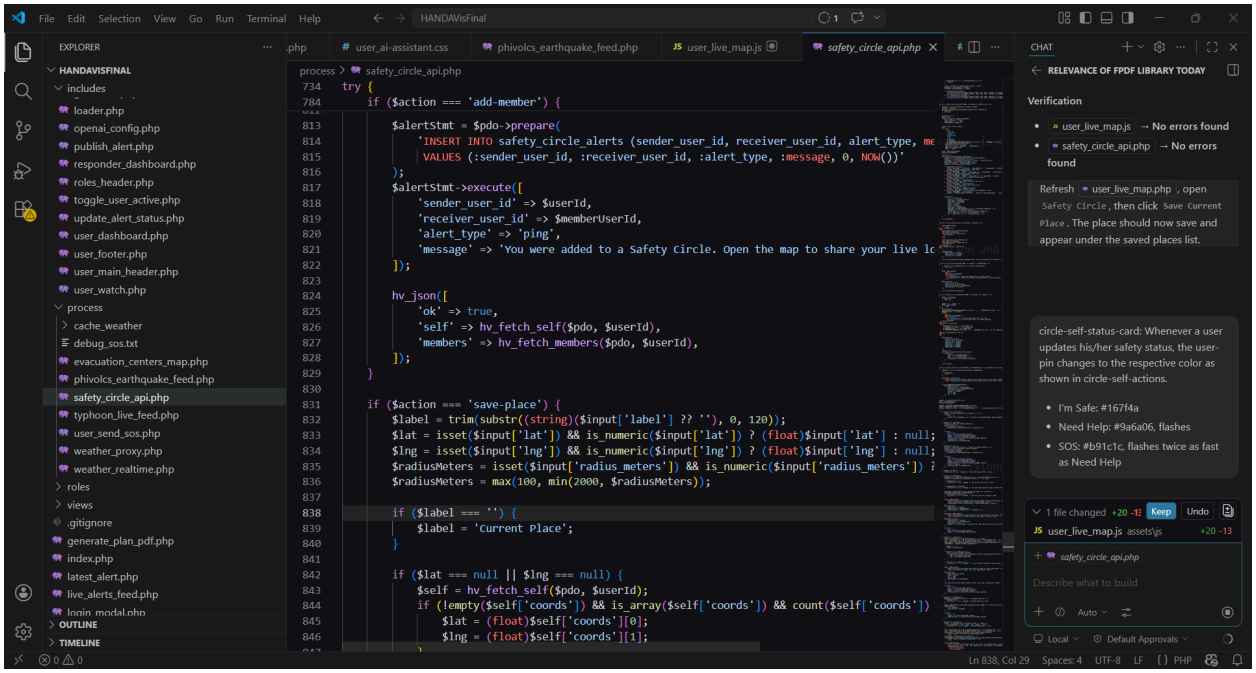
Location and Mapping Data
Geographical structure of the system.
- provinces
- municipalities
- barangays
- evacuation_centers
- road_conditions

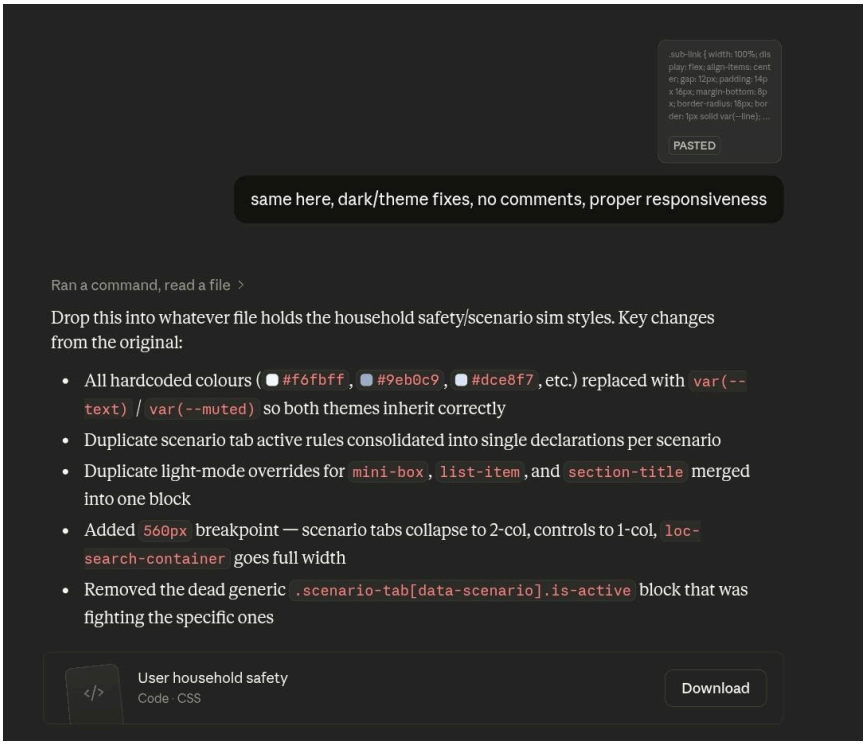
Hazard Classifications
Defines the types and severity of injuries and disasters.
- hazard_types
- hazard_type_department_map
- injury_levels
- people_affected_levels

Notification System
Handles Communication.
- notifications

Additional Data
Provides educational or awareness videos with different types of disasters for users.
- disaster_videos

System Testing, Results, and AI Integration





20:15 LTE

ChatGPT

WHAT CAN U RECOMMEND I SHOULD PUT IN THE SCENARIO SIMULATION, MAKE SURE ITS USEFUL AND FUNCTIONAL THAT USERS NEED

Thought for 29s >

Make the **Scenario Simulation** panel a **"What should I do if this happens in my area?"** tool, not just a visual demo.

What users actually need is:
prediction + impact + exact actions.

A strong setup for HANDAVis would be this:

Best structure for the panel

When a user clicks **Flood**, **Fire**, or **Storm**, open a detailed simulation view inside that box or in a modal/side panel with:

1. Scenario Inputs
Let users set:

- barangay / location

Ask anything

20:16 LTE

ChatGPT

send me the recommendation on what i should add

Thought for a couple of seconds

Add these to **Scenario Simulation** so it becomes useful, not just decorative:

What you should add

1. Location selector

- Barangay / city dropdown
- "Use my current location" button

This makes the simulation personal to the user's area.

2. Severity selector

- Low
- Moderate
- High
- Critical

Each scenario should change depending on

Ask anything

Claude

Appointment scheduling form with date and time selection

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Evacuation centers database for ...
Database connection error in PHP
Binary tree traversals and express...
Transport and application layer fu...
User account table with geographi...
SQL join syntax correction
SMS API returning 403 forbidden ...
Appointment form buttons stuck ...
Show me how compound interest ...
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```
  maxLng: 123.8
};

earthquakeReports = earthquakeReports
  .filter(function (quake) {
    // Must be within 72 hours
    if (quake.reportedAt) {
      var reportedTime = new Date(quake.reportedAt).getTime();
      if (!isNaN(reportedTime) && reportedTime < seventyTwoHoursAgo) return false;
    }

    // Must be within or near Western Visayas bounding box
    var lat = Number(quake.coords[0]);
    var lng = Number(quake.coords[1]);
    return (
      lat >= WESTERN_VISAYAS_BOUNDS.minLat && lat <= WESTERN_VISAYAS_BOUNDS.maxLat &
      lng >= WESTERN_VISAYAS_BOUNDS.minLng && lng <= WESTERN_VISAYAS_BOUNDS.maxLng
    );
  });
```

Key changes summarized:

- Popup order is now exactly: Date/Time (centered) → Epicenter → Depth → Magnitude | Intensity → damage/aftershock line (only if applicable) → tsunami line (always shown).
- Color thresholds for magnitude now follow PHIVOLCS: green for <5, yellow for 5–5.9, orange for 6–6.9, red for ≥7.
- Intensity color thresholds follow PHIVOL cale: green for I–III, yellow for IV–V, orange for VI–VII, red for VIII–X.

Reply...

Sonnet 4.6

Claude is AI and can make mistakes. Please double-check responses.

Claude

User account table with geographic and role fields

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```
WHERE `city_name` IN (
  'Bacolod','Bago','Cadiz','Escalante','Himamaylan',
  'Kabankalan','La Carlota','Sagay','San Carlos',
  'Silay','Sipalay','Talisay','Victorias'
);
```

And if you'd prefer to bake the `type` column into the original `CREATE TABLE` from the start rather than altering it after, here's the updated definition:

```sql
CREATE TABLE `tbl_city_municipality` (
  `city_id`     SMALLINT(5) UNSIGNED                NOT NULL AUTO_INCREMENT,
  `province_id` TINYINT(3) UNSIGNED                 NOT NULL,
  `city_name`   VARCHAR(100)                        NOT NULL,
  `type`        ENUM('City', 'Municipality')        NOT NULL DEFAULT 'Municipality',
  PRIMARY KEY (`city_id`),
  KEY `fk_city_province` (`province_id`),
  CONSTRAINT `fk_city_province`
    FOREIGN KEY (`province_id`)
    REFERENCES `tbl_province` (`province_id`)
    ON UPDATE CASCADE ON DELETE RESTRICT
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
```

That way the `type` is set per row right in the `INSERT` statements — no separate `UPDATE` pass needed. Want me to redo all the `INSERT` blocks with the `type` column included?

Reply...

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User account table with geographic and role fields

Share

City / Municipality

— Select city / municipality —

Barangay

— Select barangay —

USER TYPE

User | Barangay | Responder | Admin

Create account

The form is fully wired up with your Aklan data. Here's how it works:

Cascading logic — selecting a province populates the city/municipality dropdown. Selecting a city populates the barangay dropdown. Downstream selects are disabled until their parent is chosen, preventing invalid combinations.

User type — the four types are presented as clickable tiles instead of a plain dropdown, making the selection more intentional for something that controls access level.

Extending to other provinces — the `DATA` object in the script is just a JavaScript object matching the template you provided. Paste in the remaining five provinces (Antique, Capiz, Guimaras, Iloilo, Negros Occidental) in the same format and they'll appear automatically in the province dropdown. Want me to prepare the INSERT statements for those provinces once you have the data ready?

Reply...

Sonnet 4.6

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